ENPM673 Perception for Autonomous Robots (Spring 2026)
TurtleBot4 Final Project

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Logistics

1. Final demo date: May 10, 2026.
2. Location: RAL Lab.
3. Time: 10:00 AM to 2:30 PM, depending on the group. There is a 30-minute lunch break from 12:00 PM to 12:30 PM.
4. A timetable will be published on Canvas showing each group's allocated time.
5. The final project will be scored out of 100 based on demo performance.
6. The practice session schedule will be shared via Canvas within this week.

Project Description

You will implement a working intelligent robot using the TurtleBot 4 hardware provided by the Robotics Lab, gaining hands-on hardware experience and exposure to the lab's capabilities. The robot is equipped with a color camera as its primary vision sensor.

Webots Simulation Setup

You can find an example Webots simulation of the setup here: `https://github.com/adil275/ENPM673-Final-Project-Simulation.git`

Note: The final demo course layout may be different from the simulation, so you should ensure the robustness of your algorithms.

Project Design

Each task below is owned by one team member. However, team members are encouraged to collaborate and help each other. Part of the grade will be given based on overall project performance. All four tasks will run in sequence during the demo. Students should coordinate integration to ensure smooth handoffs between tasks.

You are open to using any computer vision techniques to achieve the following tasks, including machine learning, but you must defend your approach in the final demo.

Tasks

Task 1: Path Following via Arrow Signs (25 pts)

The robot navigates a marked course by detecting black arrows printed on white paper mounted along the route. Each arrow indicates the next turn direction. The robot must detect the arrow, determine which way it points, execute the corresponding turn, and proceed to the next sign until the end marker is reached. A green bounding rectangle must be overlaid on each detected arrow in real time. The layout of arrows might be different in the demo setting.

Task 2: UMD Logo Detection & Stop (25 pts)

While navigating, the robot must detect a UMD Terrapin logo placed along the route. Upon detection, it must stop completely, display a labeled red bounding box on screen, hold for 3 seconds, then resume. You are free to use any approach; a fine-tuned ML model or a classical feature-matching approach are both valid.

Task 3: Dynamic Object Detection (25 pts)

A ball will be rolled across the robot's path at an unpredictable moment during the run. The robot must detect the moving object, draw a yellow bounding box labeled "MOVING" around it, and stop until the object clears the path. Additionally, the robot must compute the time to contact (TTC) to this ball. This is the time that, if the robot continues at the same speed, it will hit the ball. When computing TTC, it is acceptable to assume translational motion only and ignore rotation.

Task 4: Continuous Horizon Detection (25 pts)

Throughout the entire run, the robot must continuously detect and overlay the horizon line on its live camera feed. The bounding rectangle from the arrow detection in Task 1 serves as the region of interest (ROI) for horizon estimation. The overlay must be visible in every frame and is evaluated from the full recorded video of the run.

Project Rules

- Bounding boxes and detection overlays must be visible live during the demo in RViz or an OpenCV window.
- Pre-trained model weights are permitted, but they must be declared in the submission report.
- Each group has a hard 30-minute slot. All setup, calibration, and runs must fit within this window.
- Only the best run counts.
- Each student must individually present their subtask during the final demo.